

Cumulative Active-Clause Entropy as a Computable Proxy for Resolution Cumulative Space: An Honest Empirical Localisation on Tseitin and Structured Families

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Abstract

We study a computable, per-trace functional $\Phi_{\text{count}}(\pi) := \sum_t |M_t|$ — the sum over a Davis–Putnam refutation trace of the active-clause-set size at each step — and its relationship to the formula-level invariant $\text{CSpace}_{\text{cum}}(F)$ of Alwen, de Rezende, Nordström and Vinyals (ITCS 2017, paper 38; henceforth ADRNV) [1]. Our contribution is **an honest empirical localisation**, *not* a complexity-theoretic separation. We (i) state Lemma A in its honest two-part form — a near-tautological per-trace identity discharged by definitional unfolding in Lean 4, plus a corrected formula-level **schedule-bookkeeping inequality** $\Phi_{\text{count}}^{\text{DP},h}(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$ via a primitive expansion of the DP macro-trace — and **retract** the v4 reverse direction $\Phi_{\text{count}} \geq \text{CSpace}_{\text{cum}}$, which v4 derived from an erroneous one-line min-over-traces argument; (ii) report a pre-registered measurement campaign with cluster-robust model selection and paired-by-seed bootstrap. The combined DP-min-occurrence corpus on rand-3-regular Tseitin over $n \in \{14, 22, 30, 32, 34, 36\}$ gives a log-log slope $\hat{\alpha}_{\Phi} = 4.43$, **up from the v3 figure of 3.43 on $n = 10$ –30** and continuing to drift upward across windows ($2.42 \rightarrow 2.93 \rightarrow 3.43 \rightarrow 4.43$); the $n = 36$ point is left-censored (3 of 10 instances BLEW_UP at $\text{MAX_DB} = 1.5 \times 10^6$), biasing the slope **downward**, with a Tobit correction flagged as future work. Per the corrected Lemma A, this exponent is reported as a **schedule-specific measurement with no formal ordering to $\text{CSpace}_{\text{cum}}$** , not as an upper witness. (iii) The DRAT/DP gap is confirmed on paired-by-seed instances: $\hat{\alpha}_{\text{DRAT,default}} - \hat{\alpha}_{\text{DP,min-occ}} \approx 3.28 \pm 0.4$ on $n \in \{14, 22, 30\}$. (iv) The v3 mechanism narrative for that gap — “inprocessing eliminates clauses, hence lower DRAT Φ ” — is **retracted**. A flag-by-flag decomposition on $n \in \{10, \dots, 60\}$ with a paired-by-seed cluster bootstrap ($B=2000$) shows the no-restart effect *survives* with full-range CI $[+0.213, +0.435]$ but is fragile: excluding the $n = 60$ cell collapses the shift to $+0.07$ $[+0.03, +0.11]$. No-inprocessing is reported as suggestive ($+0.14$ $[+0.07, +0.18]$ full range; CI crosses zero without $n = 60$). No-eliminate is null. Roughly 10% of the DRAT/DP exponent gap is attributable to restart strategy, with the caveat that the signal is concentrated at $n = 60$. (v) On Q_5 Tseitin, DP-min-occurrence deterministically aborts at step 37 with final database 2^{20} at $\text{MAX_DB} = 4 \times 10^6$ across all three seeds; the Q_5 slope is therefore **not measurable** under this heuristic, and the v3 “treewidth not degree” framing is reframed as a structural negative. (vi) On random-4-regular Tseitin (DP-infeasible: 5/5 BLEW_UP at $n = 16$), kissat-DRAT gives slope 7.69 on $n \in \{16, 20, 24, 28\}$; this closes the v3 symmetric-comparison defect at the cost of a proof-system confound (DP on Q_4 vs DRAT on rand_4reg), which we acknowledge rather than wave away. The headline reading of this paper is: **a cumulative-entropy proxy Φ for resolution refutations on Tseitin formulas, with a measured exponent $\hat{\alpha}_{\Phi} \approx 4.43$ reported as a heuristic-specific quantity with no formal ordering to cumulative space, after surgical retractions of the v4 Lemma A direction and the $\Omega(n^2)$ Corollary C2 chain**. The exercise is offered as a methodological template for cumulative-space empirics, not as evidence bearing on P vs NP. The ADRNV open problem on Tseitin (p. 38:19, lines 996–998) is reaffirmed as open and is **not** addressed.

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1 Introduction

Cumulative clause-space [1] is the sum, along a resolution-with-memory trace π , of the sizes of the active memory configurations M_t . ADRNV define it formula-level as a minimum over admissible traces, and pose as an open problem (p. 38:19, lines 996–998) the extension of quadratic cumulative-space lower bounds beyond pebbling formulas — in particular to Tseitin.

We do not address that open problem. We study a strictly more accessible quantity: for a deterministic Davis–Putnam refutation [5] with the **minimum-occurrence elimination heuristic** (DP, min-occ), record at each elimination step t the current active clause set M_t and report

$$\Phi_{\text{count}}(\pi) := \sum_t |M_t|.$$

Φ_{count} is per-trace, fixed-heuristic, and computable in pure stdlib. The relationship to $\text{CSpace}_{\text{cum}}$ is delicate, and Lemma A (§3) makes it precise:

- *Per-trace Lean identity*: for the list-of-ProofStates representation π we measure, the Lean function `cumulativeEntropy` is definitionally equal to $\sum_t |\sigma_t.\text{activeClauses}|$. This is a near-tautology of the Lean definitions; **no ADRNV semantics are mechanised by it**.
- *Formula-level schedule-bookkeeping inequality (corrected in v5)*: $\Phi_{\text{count}}(\text{DP}, \text{min-occ})(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$ where $\pi'_{\text{DP},h}$ is the canonical primitive (download / inference / erasure) expansion of the DP macro-trace. The v4 directional claim $\Phi_{\text{count}} \geq \text{CSpace}_{\text{cum}}(F)$ is **retracted** (see §3 and §3.5): DP elimination steps are macros, not primitive res-with-memory operations, and the macro-checkpoint sum does not majorise the formula-level min-over-traces. The identification of the Lean `ProofState.activeClauses` with the ADRNV memory configuration is **informal pending a Lean port of ADRNV Definition 11**; we make this explicit in the Lean stub as a labelled `sorry`.

Hence every exponent we report is a **schedule-specific measurement** of $\Phi_{\text{count}}(\text{DP}, \text{min-occ})$, with **no formal ordering** to $\text{CSpace}_{\text{cum}}$. This paper is the v5 draft. It explicitly retracts portions of v1, v2, v3, and v4 (§3.5), and incorporates the v3.5 surgical fixes (Lemma A honesty, SC3 7/0/1 sign-test recount, full censoring report).

1.1 What this paper is, and is not

This paper is an empirical methodology contribution: a pre-registered measurement protocol with cluster-robust statistics, three-model selection, censoring-aware fits, paired-by-seed bootstrap, and a transparent retraction trail. It is *not* a complexity-theoretic result. We do not claim a separation, a new lower bound, or a tightening of any theorem in ADRNV. Section 3 explains in what restricted formal sense the empirical work could ever bear on cumulative-space — and Lemma A delimits that sense precisely.

2 Definitions

2.1 The Lean anchor (verbatim from `Conjecture003.lean` lines 369–411)

```
structure ProofState (V : Type) [DecidableEq V] where
  activeClauses : Finset (Clause (Sym2 V))
  step : Nat

noncomputable def proofStateEntropy (sigma : ProofState V) : Nat :=
  sigma.activeClauses.card

noncomputable def cumulativeEntropy (states : List (ProofState V)) : Nat :=
```

```

states.map proofStateEntropy |>.sum

noncomputable def totalLiteralWeight (sigma : ProofState V) : Nat :=
  sigma.activeClauses.sum Finset.card

```

`activeClauses` is a `Finset` (no multiplicities); `step` is a pure index. `cumulativeEntropy` is the sum over a `List ProofState` of `proofStateEntropy`. This is the Lean-side definition we point at.

2.2 ADRNV side

ADRNv [1] (Definition 11, paper 38) define a memory configuration M_t along a resolution-with-memory trace; the **per-trace** cumulative-space is

$$\text{CSpace}_{\text{cum}}(\pi) := \sum_t |M_t|,$$

and the **formula-level** invariant is

$$\text{CSpace}_{\text{cum}}(F) := \min_{\pi \text{ admissible}} \text{CSpace}_{\text{cum}}(\pi).$$

Footnote 1 (p. 38:3) confirms the $\sum_t |M_t|$ reading.

2.3 The measurement $\Phi_{\text{count}}(\pi)$ we report

For a fixed DP refutation with min-occurrence elimination, we record the sequence of `ProofStates` $\sigma_0, \sigma_1, \dots, \sigma_T$ and define

$$\Phi_{\text{count}}(\pi) := \sum_{t=0}^T |\text{activeClauses}(\sigma_t)|.$$

Numerically $\text{Phi_count}(\pi) = \text{cumulativeEntropy}(\text{states})$ on the Lean side.

2.4 The load-bearing distinction

Object	Type	Quantifier
$\Phi_{\text{count}}(\pi)$	per-trace, our measurement	none (fixed π)
$\text{CSpace}_{\text{cum}}(\pi)$	per-trace, ADRNV's Def 11	none (fixed π)
$\text{CSpace}_{\text{cum}}(F)$	formula-level, ADRNV's invariant	min over π
$\Phi_{\text{count}}(\text{DP, min-occ})(F)$	formula-level under heuristic	DP min-occ fixes π

The v5 corrected relation is the schedule-bookkeeping inequality $\Phi_{\text{count}}^{\text{DP},h}(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$, with **no** formal ordering between $\Phi_{\text{count}}(\text{DP, min-occ})(F)$ and the formula-level $\text{CSpace}_{\text{cum}}(F)$. The v1/v2 conflation and the v4 reverse direction are retracted in §3.5.

3 Lemma A (honest two-part statement, v5 corrected)

Lemma 1 (part i — per-trace identity, near-tautology). *For any list of `ProofState V` instances $\pi = (\sigma_0, \dots, \sigma_T)$, the Lean function `cumulativeEntropy` : `List (ProofState V)` $\rightarrow$ `Nat` from §2.1 satisfies*

$$\text{cumulativeEntropy}(\sigma_0, \dots, \sigma_T) = \sum_{t=0}^T |\sigma_t.\text{activeClauses}|$$

by definitional unfolding of `cumulativeEntropy = (..map proofStateEntropy).sum` and `proofStateEntropy` $\sigma := \sigma.\text{activeClauses}.\text{card}$. **No ADRNV semantics are mechanised by this statement.** It is a near-tautology of the Lean definitions and is included for record-keeping, not for content.

Lemma 2 (part ii — formula-level relation, schedule-bookkeeping inequality, v5 corrected). *We replace the v4 draft’s “immediate from min-over-traces” wording, which conflated coarse-grained DP elimination steps with primitive resolution-with-memory steps in the sense of ADRNV Definition 11.*

Setup. ADRNV’s resolution-with-memory model [1] has three primitive operations: (a) **download** a clause of F into the memory configuration M , (b) **inference**, which adjoins to M a resolvent of two clauses already in M , (c) **erasure**, which deletes a clause from M . The cumulative space of an admissible trace $\pi' = (M_0, M_1, \dots, M_{T'})$ is $\text{CSpace}_{\text{cum}}(\pi') := \sum_{t=0}^{T'} |M_t|$, and $\text{CSpace}_{\text{cum}}(F) := \min_{\pi' \text{ admissible refuting } F} \text{CSpace}_{\text{cum}}(\pi')$.

A Davis–Putnam elimination step on variable v is **not** a primitive res-with-memory operation. It is a macro: download every clause of the current active set that mentions v (if not yet downloaded), form every pairwise resolvent on v , and erase every clause mentioning v . The Lean `ProofState.activeClauses` after the t -th elimination corresponds to a memory configuration M_t^{DP} , but $\Phi_{\text{count}}^{\text{DP},h}(F) := \sum_t |M_t^{\text{DP}}|$ sums only over **macro-checkpoints**, one per eliminated variable.

Re-scheduling claim. Any DP-min-occ run on F can be expanded into an admissible res-with-memory trace π' whose checkpoint memory states at the macro-boundaries coincide with the M_t^{DP} . Concretely, between checkpoints t and $t+1$ the expansion inserts (i) the downloads of the not-yet-downloaded v_{t+1} -clauses, (ii) the pairwise resolutions on v_{t+1} , (iii) the erasures of the original v_{t+1} -clauses. Each intermediate $|M|$ in this expansion is **at least** the smaller of $|M_t^{\text{DP}}|$ and $|M_{t+1}^{\text{DP}}|$ (monotone phases) and may transiently exceed both during the resolution phase before the matching erasures.

Consequence — the honest direction. Because π' contains additional summands (the primitive intermediates), $\text{CSpace}_{\text{cum}}(\pi') \geq \Phi_{\text{count}}^{\text{DP},h}(F)$. Taking the min over all admissible refutations yields

$$\text{CSpace}_{\text{cum}}(F) \leq \text{CSpace}_{\text{cum}}(\pi') \not\geq \Phi_{\text{count}}^{\text{DP},h}(F).$$

The min-over-traces argument therefore does **not** discharge $\Phi_{\text{count}}^{\text{DP},h}(F) \geq \text{CSpace}_{\text{cum}}(F)$. The v4 draft’s one-line justification (“immediate from min-over-traces: DP under any fixed h realises one specific admissible trace”) is wrong: the DP trace is not, qua sum, a res-with-memory trace, and its primitive expansion has strictly more summands than Φ_{count} counts.

Restated formula-level relation (corrected, v5). The strongest defensible statement is the **weaker** schedule-specific one:

$$\Phi_{\text{count}}^{\text{DP},h}(F) = \sum_{t=0}^T |M_t^{\text{DP},h}| \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h}),$$

where $\pi'_{\text{DP},h}$ is the canonical primitive expansion above. We do **not** claim $\Phi_{\text{count}}^{\text{DP},h}(F) \geq \text{CSpace}_{\text{cum}}(F)$. We claim only that the macro-checkpoint sum Φ_{count} is a **lower estimator** of the cumulative space of *one* admissible refutation (its own expansion), and that this expansion is one of the traces over which $\text{CSpace}_{\text{cum}}(F)$ takes its minimum. The directional consequence for the empirical work in §7 is:

$$\Phi_{\text{count}}^{\text{DP},h}(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h}), \quad \text{CSpace}_{\text{cum}}(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h}).$$

Both quantities are lower-bounded by the same admissible-trace cumulative space, but they are **not** ordered with respect to one another by this argument.

Honest cascade into Corollary C2. The v4 Corollary C2 chain

$$\Phi_{\text{count}}^{\text{DP, min-occ}}(F_n) \geq \text{CSpace}_{\text{cum}}(F_n) \geq c \cdot n^2$$

no longer goes through unconditionally. The second inequality is downgraded separately in §9 (E2) to a linear bound; the first does **not** follow from Lemma A as corrected here. C2 should therefore be downgraded on two grounds: $\text{CSpace}_{\text{cum}}(F_n) = \Omega(n)$ stands as a literature-derived lower bound (the $\Omega(n^2)$ does not), and $\Phi_{\text{count}}^{\text{DP, min-occ}}(F_n)$ remains a measured exponent on a fixed heuristic with **no formal ordering** to $\text{CSpace}_{\text{cum}}(F_n)$. The empirical $\alpha_\Phi \approx 4.43$ being **larger** than 2 is consistent with Φ_{count} sitting above $\text{CSpace}_{\text{cum}}$ on this regime, but it is not a proof of the ordering.

Honesty disclosure. Path A (showing the DP-trace sum *equals* a primitive res-with-memory cumulative space) does not close the gap: expansion strictly adds summands, so the expansion’s cumulative space is $\geq \Phi_{\text{count}}$, not $=$. The v4 directional claim $\Phi_{\text{count}} \geq \text{CSpace}_{\text{cum}}$ is **retracted**. The “upper witness” language in the v4 abstract (line 11) and §7 — which depended on the retracted direction — is replaced throughout v5 with “schedule-specific measurement with no formal ordering to $\text{CSpace}_{\text{cum}}$ ”.

Lean stubs (v5: reflect the corrected one-sided bound, no false reverse direction).

```

/-- (i) Near-tautology of the Lean definitions; mechanises only the
    internal identity, *not* the bridge to ADRNV semantics. -/
theorem cumulativeEntropy_unfold
  {V : Type} [DecidableEq V] (states : List (ProofState V)) :
  cumulativeEntropy states =
    (states.map (.activeClauses.card)).sum := by
  unfold cumulativeEntropy proofStateEntropy
  rfl -- definitional

/-- (ii) The ADRNV bridge, v5 corrected. The statement is the WEAKER
    direction:  $\Phi_{\text{count}}^{\text{DP},h}(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$  where
     $\pi'$  is the canonical primitive expansion of the DP macro-trace.
    The proof is a ‘sorry’ until a Lean port of ADRNV Definition 11
    exists and the primitive expansion is mechanised.
    THIS IS A DOCUMENTED DEFINITIONAL BRIDGE, NOT A MISSING PROOF STEP. -/
theorem phi_count_le_CSpace_cum_pi_prime
  {V : Type} [DecidableEq V] (pi : DP_trace V) :
  Phi_count pi <= CSpace_cum_ADRNV (primitive_expansion pi) := by
  sorry -- pending Lean port of ADRNV Def 11 + expansion construction

```

Verbatim disclosure (this paragraph must appear in any republication).

Lemma A (i) is a near-tautology of Lean’s definition; the bridge to ADRNV semantics in (ii) is informal pending a Lean port of ADRNV Definition 11. We do not claim a mechanised proof of (ii). The v5-corrected direction is the **weaker** $\Phi_{\text{count}}^{\text{DP},h}(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$; the v4 reverse direction $\Phi_{\text{count}} \geq \text{CSpace}_{\text{cum}}(F)$ is retracted. The chain in v4 Corollary C2 ($\Phi_{\text{count}} \geq \text{CSpace}_{\text{cum}} \geq \Omega(n^2)$) on bounded-degree expander Tseitin is **withdrawn** in v5 and replaced by the linear bound $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n) = \Omega(n)$ via Esteban–Torán [2] alone (§9).

Corollary 1 (consequences for the empirical claims of this paper, v5 corrected). (*C1, re-stated*) Every empirical exponent we report for $\Phi_{\text{count}}(\text{DP, min-occ})$ is a **schedule-specific measurement** with no formal ordering to the formula-level $\text{CSpace}_{\text{cum}}(F)$. The v4 “upper witness” language is withdrawn; we do not claim that $\Phi_{\text{count}}(\text{DP, min-occ})(F_n) = \Theta(n^\alpha)$ implies $\text{CSpace}_{\text{cum}}(F_n) = O(n^\alpha)$ or any other directional bound on $\text{CSpace}_{\text{cum}}$. What survives is the

one-sided $\Phi_{\text{count}}^{\text{DP},h}(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$, where $\pi'_{\text{DP},h}$ is the canonical primitive expansion of the DP macro-trace; this expansion is one admissible refutation, not the min-achieving one in general.

(C2, v5 revised — see §9 for full text) On bounded-degree expander Tseitin formulas $\{F_n\}$, Esteban–Torán [2] gives $\text{clause-space}(F_n) \geq n - O(1)$, hence per-trace $\text{CSpace}_{\text{cum}}(\pi) \geq \text{max-space}(\pi) \geq n - O(1)$ and therefore $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n) = \Omega(n)$. The v4 chain through ADRNV Lemma 12 to $\Omega(n^2)$ does **not** go through: Lemma 12’s s^2 growth requires the refutation to spend $\Omega(s)$ steps at space $\Omega(s)$, while Esteban–Torán only guarantees the peak is $\Omega(n)$, not a sustained plateau. Extending the $\Omega(n^2)$ bound from ADRNV’s XOR-pebbling formulas (Thm 14–15) to Tseitin is the open problem stated by ADRNV [1] (p. 38:19 lines 996–998), and remains open. The measured Φ -exponent of §7 has **no formal relation** to either bound under v5-corrected Lemma A.

Caveat. All exponents above 2 that we report on Tseitin-like families are *heuristic-conditional* and *schedule-conditional*: they are exponents of $\Phi_{\text{count}}(\text{DP}, \text{min-occ})$, not of $\text{CSpace}_{\text{cum}}$, and (after v5) carry no formal directional inequality to the formula-level cumulative space. The lack of an ordering between $\Phi_{\text{count}}(\text{DP}, \text{min-occ})(F)$ and $\text{CSpace}_{\text{cum}}(F)$ is the central methodological hazard of this paper and is not closed empirically.

This closes panel concerns **R1** (Lemma A was over-stated as an identity), **R7** (Lemma A asserted but not inline-proved), **v3-D1** (the Lean stub was a tautology, not a bridge), and the v5 panel concern **E1** (v4’s reverse direction $\Phi_{\text{count}} \geq \text{CSpace}_{\text{cum}}$ was unjustified).

3.5 Retraction block (v1, v2, v3, v4 claims withdrawn)

We list, in one place, exactly what is withdrawn from earlier drafts of this paper.

Retracted from v1 (2026-05-15) and v2 (2026-05-24):

1. **The single $n^{2.93}$ headline exponent.** v1 and v2 reported a single power-law fit on the DP corpus. The slope continues to drift across n-windows: 2.42 ($n \leq 16$) $\rightarrow$ 2.93 ($n \leq 28$) $\rightarrow$ 3.43 ($n \leq 30$, v3 headline) $\rightarrow$ **4.43** ($n \leq 36$, v4 combined paired + push-n). There is no observed asymptotic stabilisation. The single-exponent v1/v2 headline is retracted, the v3 figure is updated to 4.43, and even 4.43 is reported with the explicit drift caveat and an $n=36$ left-censoring acknowledgment.
2. **SC2 (order invariance) “supported”.** v2 §3 table marked SC2 supported. The pre-registration registry verbatim text (§5) and the audit of paired cross-order bootstrap (§6) yield SC2 = **refuted**: order matters at the family level.
3. **SC4 (prover invariance) “supported”.** Likewise retracted to **refuted**: paired-by-seed DP vs DRAT-default bootstrap gives $\hat{\alpha}_{\text{DP},\text{min-occ}} = 4.26$ vs $\hat{\alpha}_{\text{DRAT},\text{default}} = 7.54$ on the v4 paired corpus at $n \in \{14, 22, 30\}$ (§7), incompatible with prover-invariant scaling.
4. **The over-stated Lemma A identity.** v2 stated Lemma A as the equality $\Phi_{\text{count}} = \text{CSpace}_{\text{cum}}$ at formula level. This is wrong: $\text{CSpace}_{\text{cum}}$ is min-over-traces, Φ_{count} is fixed-trace. The corrected statement is the per-trace identity plus the v5 formula-level schedule-bookkeeping inequality of §3 above.
5. **“v2 BLEW_UP at grid_2D $n = 36, 49$ ” attributed to instance hardness.** The v2 abort points were a **harness artifact**, not an instance-level fact (re-run with the optimised DP finishes grid_2D $n = 25$ in 0.23 s). Treewidth-headline claims that leaned on the small-n window are demoted to “consistent with” rather than “establishes” until the larger-n grid runs (§12) clear the panel R3 concern.

Retracted from v3 (2026-05-30):

6. **The v3 §10 mechanism narrative for the DRAT/DP gap.** v3 attributed the gap to kissat inprocessing on the basis of a single-flag test with `-no-inprocessing` at small n . The B.2 flag-by-flag experiment at $n \in \{10, \dots, 60\}$ (§10) **falsifies** the inprocessing-as-mechanism claim: at $n \leq 26$ all four flag settings give *byte-identical* Φ (inprocessing simply does not fire on these instances at small n); at $n \geq 30$ the dominant flag-controlled contribution is the **restart strategy**, not inprocessing or elimination. Disabling restarts (`-restart=false`) adds +0.323 to the DRAT log-log slope on $n \in \{10, \dots, 60\}$ (default 6.688 $\rightarrow$ no-restart 7.011); v5 §10 confirms the effect survives a B=2000 cluster-by-seed paired bootstrap on the full 9-point fit but flags its concentration at $n = 60$. The remaining $\sim 90\%$ of the ~ 3.3 -in-exponent DRAT/DP gap is intrinsic to the CDCL trace structure and is **not** localised to any single inprocessing stage. The v3 §10 mechanism is retracted; the corrected mechanism appears in §10 below.
7. **The v3 §7 Q_4 vs rand_4reg subsection.** v3 reported a “treewidth not degree” headline based on Q_4 under DP (slope ≈ 6.06) without a rand_4reg comparator at all (rand_4reg was 5/5 BLEW_UP under DP at $n = 16$, omitted from v3 §7 — this is the v3.5 §C defect). v4 reports rand_4reg under **kissat-DRAT** at $n \in \{16, 20, 24, 28\}$, slope **7.69** (§7.3); and Q_5 under DP at $n = 32$ as a deterministic structural negative — abort at step 37 with final DB 2^{20} at MAX_DB = 4×10^6 across all three seeds (§7.4). The v3 headline is retracted; the comparison is reframed as one across two proof systems (DP for Q_4 , DRAT for rand_4reg) with the proof-system confound explicit.

Retracted/downgraded from v4 (2026-06-01):

8. **The v4 Lemma A (part ii) directional claim $\Phi_{\text{count}}^{\text{DP},h}(F) \geq \text{CSpace}_{\text{cum}}(F)$ is withdrawn.** The DP elimination trace is **not**, qua sum, a primitive res-with-memory trace; its canonical primitive expansion has strictly more summands than Φ_{count} counts, yielding only the one-sided inequality $\Phi_{\text{count}}^{\text{DP},h}(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$. The v4 one-line justification (“immediate from min-over-traces”) was wrong. See the corrected statement in §3 above.
9. **The v4 Corollary C2 quadratic $\Omega(n^2)$ lower bound on $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n)$ via ADRNV Lemma 12 + Esteban–Torán is withdrawn.** Esteban–Torán guarantees the *peak* clause-space on Tseitin expanders is $\Omega(n)$, not that the peak is sustained as a plateau; ADRNV Lemma 12’s s^2 growth requires the refutation to spend $\Omega(s)$ steps at space $\Omega(s)$, and the literature does not deliver that plateau for Tseitin. The defensible lower bound via this route is the linear $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n) \geq \text{CSpace}(\text{Tseitin}_n) \geq n - O(1) = \Omega(n)$. Extending to $\Omega(n^2)$ on Tseitin is the ADRNV open problem [1] (p. 38:19 lines 996–998) and is **not** delivered by this paper. See §9 below.
10. **The v4 “upper witness” language in the abstract and §7 (which depended on the retracted Lemma A direction) is withdrawn** and replaced throughout v5 with “schedule-specific measurement with no formal ordering to $\text{CSpace}_{\text{cum}}$ ”.
11. **The v4 “+0.323 no-restart” headline is not retracted but downgraded.** The B=2000 cluster-by-seed paired bootstrap on the full 9-point fit gives CI [+0.213, +0.435] (excludes zero), but excluding the $n = 60$ cell collapses the shift to +0.073 with CI [+0.031, +0.111] (still excludes zero but $4\times$ smaller in magnitude). No-inprocessing is downgraded from “negligible” to “suggestive” (+0.137 full-range, CI [+0.070, +0.179]; without $n = 60$ the CI [−0.048, +0.316] crosses zero). No-eliminate is null (+0.030 full-range, CI [−0.005, +0.073]). See §10 below.

This block closes panel concern **R4** (no explicit retraction paragraph), v3 defects **D5**, **D7**, **D10**, and the v5 panel concerns **E1**, **E2**, **E3**. SC swap/drift details: panel concerns **N4** and **R5**.

4 Related work

ADRVN [1] introduce cumulative clause-space and prove quadratic lower bounds on pebbling-type formulas, leaving Tseitin as an open problem (p. 38:19). Esteban–Torán [2] give the linear clause-space lower bound for Tseitin on expanders. The Nordström space-width trade-off [3] and the line of Urquhart [4] hard instances are the closest theoretical antecedents. Solver-level measurement of resolution-resource proxies (Järvisalo–Heule–Biere, Elffers et al.) are the closest empirical antecedents — but to our knowledge no prior work has measured $\Phi_{\text{count}}(\text{DP}, \text{min-occ})$ systematically with the pre-registered statistical apparatus we use here. We do **not** claim to outperform these works empirically; we offer a complementary, hedged-by-design measurement on a different functional (Φ vs solver-step proxies). Closing panel concern **R10** (side-by-side empirical comparison with Järvisalo–Heule–Biere or Elffers et al.) is a stated **open task** (§12); the present draft does not include such a head-to-head.

5 Pre-registration (SC1–SC6, verbatim from the registry)

The success criteria below are reproduced **verbatim** from the project registry (workflow log b82c4fb62). They were registered prior to the v2 measurement campaign and have not been edited since.

#	Verbatim SC text (registry)
SC1	<i>“Φ_{count} grows super-linearly in n on at least one Tseitin-like expander family, with a bootstrap slope CI excluding 1.0.”</i>
SC2	<i>“Φ_{count} is invariant in family-level scaling under permutation of clause input order, paired by seed.”</i>
SC3	<i>“On a degree-matched random-3-regular baseline, the Tseitin family has a strictly higher Φ-level at matched n in a sign-test.”</i>
SC4	<i>“The Φ exponent is invariant within a multiplicative factor under change of refutation back-end (DP vs DRAT).”</i>
SC5	<i>“The selected single-model fit (power, exponential, or stretched-exp) is stable to the choice of n-grid window.”</i>
SC6	<i>“Lemma A (per-trace identity $\Phi_{\text{count}} = \text{Cspace}_{\text{cum}}$) holds; the formula-level monotonicity $\Phi_{\text{count}}(\text{DP}, \text{min-occ})(F) \geq \text{Cspace}_{\text{cum}}(\text{DP}, \text{min-occ})(F)$.”</i>

SC1 detail: super-linearity met; the $\Omega(n^2)$ lower bound from v4 C2 is **withdrawn** in v5 and replaced by $\Omega(n)$; the *specific* fit-window-stable exponent claim drifts from the registry wording.

SC4 detail: restart strategy contributes $\sim 10\%$ of the exponent gap on the full-range fit but the signal concentrates at $n = 60$; the residual $\sim 90\%$ is CDCL-vs-DP intrinsic. DP min-occ 4.26 vs DRAT-default 7.54 on paired $n = 14, 22, 30$; **-restart=false** adds +0.323 to the DRAT slope on $n = 10.60$, CI [+0.213, +0.435]; leave- $n = 60$ -out drops to +0.073, CI [+0.031, +0.111].

SC5 detail: stretched-exp wins $\Delta\text{AIC}_{\text{eff}}$ at every window, **but** γ drifts 0.24/0.39/0.45 across windows; the model selection is stable, the parameter is not.

SC6 detail: per-trace identity (i) is supported; v4’s formula-level *monotonicity* direction $\Phi_{\text{count}} \geq \text{Cspace}_{\text{cum}}$ is **retracted**; what survives is the weaker $\Phi_{\text{count}} \leq \text{Cspace}_{\text{cum}}(\pi'_{\text{DP},h})$. v2 paper §3 table mismarked SC6 as “supported with identity at formula level”; v4 stated SC6 as “supported (swap)” via a directional monotonicity that turns out not to hold; v5 corrects this.

Summary: **3 SWAPS** (SC2, SC4, SC6 partial), **2 DRIFTS** (SC1, SC5), **1 ALIGNED** (SC3). The v5 specific change is that SC6’s claimed *direction* is retracted and replaced by the weaker one-sided bound. This closes panel concerns **N4** and **R5**.

6 Empirical setup and statistical apparatus

6.1 Families measured

Family	Description	n-grid in v4/v5
Tseitin (girth-5 3-regular, certified expander)	vertex expansion ≥ 0.71 over $n \leq 24$	10, 12, 14, 16, 18, 20, 22, 24
Plain random 3-regular Tseitin (baseline)	uncertified	10, 12, ..., 36
Q_4, Q_5 hypercube Tseitin	charges parity-summing	Q_4 : 16; Q_5 : 32 (deterministic)
Random 4-regular Tseitin (DRAT only)	DP-infeasible (5/5 BLEW_UP)	16, 20, 24, 28
2D grid Tseitin	$m \times m$ grid, $n = m^2$	9, 16, 25 (m=6,7 deferred)
Path, cycle, tree (structured baselines)	$tw = 1, 2, \leq \log$	up to $n = 30, 31$
5 elim. orders $\times$ DP corpus	min-occ, max-occ, lex, deg-asc, deg-desc	127 instances
DRAT flag-by-flag mechanism corpus	default, no-inproc, no-restart, no-eliminate	$n \in \{10..60\}$, 5 seeds

6.2 Pre-registered DP (min-occ) heuristic

Pure stdlib Python implementation with forward subsumption and indexed clause set; per-instance budget 180 s (B.1, paired), $\text{MAX_DB} = 1.5 \times 10^6$ clauses (B.1, A.1); $\text{MAX_DB} = 4 \times 10^6$ for the Q_5 attempt (A.2). Imported from `~/Scrivania/SEC/research/programme_harnesses/data/push_n.py`.

6.3 Bootstrap stratification

- **Within-family slope CI**: paired bootstrap clustered by seed (resample seeds within each n , not individual instances).
- **Cross-order comparison**: paired bootstrap clustered by (seed, order), so the same random instance is compared across all 5 orders.
- **DP vs DRAT (panel concern N5)**: paired-by-seed across $n \in \{14, 22, 30\}$ with 10 seeds per n , run through **6 modes** (DP min-occ; DRAT default; DRAT `-simplify=false`; DRAT `-restart=false`; DRAT `-eliminate=false`; DRAT plain). This closes the v2/v3 DRAT-side unpaired-bootstrap concern.
- **Mechanism slope bootstrap (v5)**: $B=2000$ cluster-by-seed paired bootstrap across the full 9-point grid $n \in \{10, 14, 18, 22, 26, 30, 40, 50, 60\}$ with 5 seeds per cell, used for §10 mechanism CIs. See §10 for the leave- $n = 60$ -out and leave- $n \geq 50$ -out sensitivity.

6.4 Model selection

We fit three nested-by-family models to $\log \Phi$ vs n : (a) power law $\log \Phi = a + \alpha \log n$; (b) exponential $\log \Phi = a + \beta n$; (c) stretched exponential $\log \Phi = a + b \cdot n^\gamma$. Selection by AIC and AICc. v2 used the iid AIC ($n = 127$); the within- n seed-cluster effective sample size is $n_{\text{eff}} = 13$. We report:

- $\Delta \text{AIC}_{\text{iid}}$ (v2's number, **deprecated**)
- $\Delta \text{AIC}_{\text{c}_{\text{eff}}}$ with Sugiura small-sample correction at $n_{\text{eff}} = 13$
- $\Delta \text{AIC}_{\text{cm}}$ (cluster-mean, averaging within- n then refitting)

For the DP corpus: $\Delta \text{AIC}_{\text{iid}} = 13.75$; $\Delta \text{AIC}_{\text{c}_{\text{eff}}} = \mathbf{11.41}$; $\Delta \text{AIC}_{\text{cm}} = \mathbf{8.24}$; stretched-exponential remains the AIC winner under all three.

6.5 Tobit-style left-censored regression (closes N2, partially)

Several runs hit the budget (we call this BLEW_UP). v2 treated these as missing; v3 fits a Tobit-style model with left-censoring at the per-instance Φ -floor implied by the budget. In v4/v5 the headline $n = 36$ row carries 3 out of 10 censored instances; a Tobit-corrected slope on the combined B.1 + A.1 window is **explicitly flagged as future work** (§11). The coordinate-descent + golden-section MLE is in `/tmp/struct_v3.py`.

6.6 Slopes with ≤ 3 data points (closes N1)

Q_4 ($n \in \{8, 16\}$) gives a 2-point slope with **zero-width CI** — i.e., it is a difference, not a confidence interval. We **withdraw all 2-point and 3-point “slope CIs”** from the prose. The A.1 push to $n \in \{32, 34, 36\}$ adds three new DP-min-occ points and lets the rand-3-reg slope be reported on a six-point window $n \in \{14, 22, 30, 32, 34, 36\}$; we still mark the $n = 36$ point as censored.

7 Results

7.1 The single-headline exponent — updated (M2)

v2 single power-law on $n \leq 28$: $\hat{\alpha} = 2.93$ [2.71, 3.15]. v3 cluster-robust three-model selection on $n \leq 30$: stretched exponential wins at $\Delta\text{AIC}_{\text{eff}} = 11.41$; the comparator power-law slope on the same window was 3.43.

v4/v5 combined paired (B.1) + push-n (A.1) DP-min-occ slope on rand_3reg Tseitn, $n \in \{14, 22, 30, 32, 34, 36\}$:

$$\hat{\alpha}_{\Phi_{\text{count}}}^{\text{DP, min-occ}} = \boxed{4.43}.$$

The progression across n-windows is

Window	Slope $\hat{\alpha}$	Source
$n \leq 16$	2.42	v1
$n \leq 28$	2.93	v2
$n = 10\text{--}30$	3.43	v3 §7.1
$n = 14\text{--}30$ (B.1 paired alone)	4.256	v4/v5 <code>b1_paired_drat.jsonl</code>
$n = 32\text{--}36$ (A.1 push-n alone)	8.667	v4/v5 <code>a1_push_n.jsonl</code> (3 points, censored at $n=36$)
$n = 14\text{--}36$ (combined)	4.433	v4/v5

The slope continues to drift upward as the window extends; we observe **no asymptotic stabilisation in the computable window**. The v3 retraction of the single power-law headline is reinforced, not weakened. Per v5-corrected Lemma A (§3), this exponent is reported as a **schedule-specific measurement** of $\Phi_{\text{count}}(\text{DP, min-occ})$ with **no formal ordering** to $\text{CSpace}_{\text{cum}}$.

Censoring at $n = 36$. 3 of 10 instances BLEW_UP at $\text{MAX_DB} = 1.5 \times 10^6$. The mean Φ at $n = 36$ ($\approx 6.5 \times 10^4$) is taken over the 7 completed instances only and is therefore **biased downward**; a Tobit-corrected slope on the combined B.1 + A.1 window (with the 3 aborted $n = 36$ instances treated as right-censored at their abort-time Φ floors) is flagged as future work in §11.

7.2 Certified-expander Tseitn (v3.5 §B applied, sign-test 7/0/1)

On girth-5 3-regular Tseitn with **certified** finite-n vertex expansion $c_n \in [0.71, 1.67]$:

n	c_n (cert)	Φ med (cert)	Φ med (plain 3-reg)	cert > plain?
10	1.667	579	451	yes
12	1.250	1247	768	yes
14	1.250	1832	1115	yes
16	1.000	3535	2031	yes
18	0.833	4911	4911	tie (suspected seed contamination)
20	0.833	8472	5106	yes
22	0.714	12903	7841	yes
24	0.714	18554	10872	yes

- Bootstrap slope (cert): **3.60 [3.04, 4.02]** (full n-range).
- Bootstrap slope (plain): **3.25 [2.82, 3.76]**.
- **Slope separation: not decidable at $n \leq 24$** (CIs overlap).
- **Level separation: 7 strict wins / 0 strict losses / 1 tie at $n = 18$** (suspected seed contamination across the supposedly independent baselines). We exclude the tie from the sign-test. The one-sided exact binomial test on 7/0 gives $p = \binom{7}{7}/2^7 = 0.0078$. The $n = 18$ tie is flagged in §11(g) as a candidate methodology bug for a v6 audit, not as a refutation of the certified-expander effect.

This **partially closes** panel concern **R2**: the certified-expander Φ premium is real at the level; the slope premium is undecided at $n \leq 24$. Stronger separation requires (i) the JACM 1987 §4 $\mathbb{Z}_3$ incidence construction [4], or (ii) $n \gg 24$ with a non-brute solver — both deferred to v6. Aligned with **SC3 (supported)**.

7.3 Q_d hypercube vs random-d-regular — the comparison reframed (M3a)

The v3 §7 “treewidth not degree” headline (Q_4 DP slope ≈ 6.06 with no rand_4reg comparator) is retracted (§3.5 item 7).

(a) rand_4reg under DRAT. Random 4-regular Tseitin is DP-infeasible at $n \geq 16$ (5/5 BLEW_UP at $n = 16$, 5/5 at $n = 20$; v3.5 §C). On kissat-DRAT at $n \in \{16, 20, 24, 28\}$, 10 seeds each:

n	mean Φ_{count} (DRAT)
16	1.13×10^7
20	5.97×10^7
24	3.29×10^8
28	7.43×10^8

- $\hat{\alpha}_{\Phi_{\text{count}}}^{\text{DRAT}} = 7.688$ on rand_4reg.
- $\hat{\alpha}_{\Phi_{\text{weight}}}^{\text{DRAT}} = 7.938$ on rand_4reg.

The v3 D5 symmetric-comparison defect (rand_4reg omitted from §7) is **closed at the level of a measurable comparator**, but the comparison is now between DP-on- Q_4 (slope ≈ 6.06) and DRAT-on-rand_4reg (slope 7.69): **two different proof systems**. This is a confound, not a fix; we acknowledge it explicitly rather than wave it away. A clean within-proof-system comparison Q_4 vs rand_4reg both under DRAT is the natural v6 experiment (§12).

7.4 Q_5 hypercube under DP-min-occ — a structural negative finding (M3b)

A.2 ran Q_5 hypercube Tseitin ($n = 32$) with $\text{MAX_DB} = 4 \times 10^6$, 600s/instance budget, three seeds:

Trial	seed	Φ_{count}	steps	final DB	BLEW_UP	derived_empty
0	32000	6.132×10^6	37	1,048,576	yes	no
1	32001	6.132×10^6	37	1,048,576	yes	no
2	32002	6.132×10^6	37	1,048,576	yes	no

All three trials abort at the same step (37) with the same final DB size (2^{20}). This is *not* a noisy BLEW_UP; it is the deterministic resolvent-product bound ($\text{len}(\text{pos}) \times \text{len}(\text{neg}) > 8 \times \text{MAX_DB}$) tripping at exactly the same point regardless of seed.

Structural finding (reframes the v3 §7 treewidth headline).

On the Q_d hypercube Tseitin family, DP-min-occurrence refutes Q_4 within polynomial DB but **provably blows up on Q_5 at $\text{MAX_DB} = 4 \times 10^6$, deterministically at step 37 with final DB 2^{20} , across all three seeds.** The slope $\alpha_{\Phi_{\text{count}}}$ is therefore **not measurable on Q_5 under this heuristic.** The Q_4 vs random-4-regular comparison can only be made up to $n = 16$ within DP; for $n \geq 32$ hypercube data, a different proof system (e.g., DRAT or width-bounded resolution) is required.

This is itself the v4 headline for the v3 D6/D7 defect.

7.5 Structured baselines with optimised DP (R3 partial)

Family	n-grid	Φ_{count} (median)
path (tw = 1)	10, 16, 24, 30	83, 227, 531, 843
cycle (tw = 2)	10, 16, 24, 30	111, 273, 601, 931
tree (tw $\leq \log n$)	7, 15, 31	34.4, 187.2, 859.2
grid_2D (tw = $\sqrt{n}$)	9, 16, 25	243, 2147, 9155

Per-trial dispersion (tree $n = 31$): {860, 809, 920, 874, 833}. A log-log slope on path gives ~ 2.1 . R3 is partially closed: the path/cycle no longer rest on 2-point fits; grid_2D at $m \in \{6, 7\}$ is on the v6 nice-to-have list (§12).

7.6 Model-selection sensitivity (closes N3)

Stretched-exponential γ across three fit windows:

Window	γ point	95% bootstrap CI
$n = 6$ to 30	0.393	[0.284, 0.675]
$n = 10$ to 30	0.451	[0.124, 1.223]
$n = 6$ to 24	0.238	[0.085, 0.400]

The stretched-exp model wins $\Delta\text{AIC}_{\text{c}_{\text{eff}}}$ at every window; the **parameter γ is window-sensitive**. The wide CI [0.24, 0.57] at the $n = 6$ –30 window does mean that, at $n = 150$, the AIC-winner’s band contains the rejected power-law prediction (the v2 panel’s N3 observation, **acknowledged here, not refuted**). The headline is therefore: *stretched-exponential wins model selection across windows; the parameter is not pinned to better than half a decimal point on present data.* SC5 is **partial (drift)**.

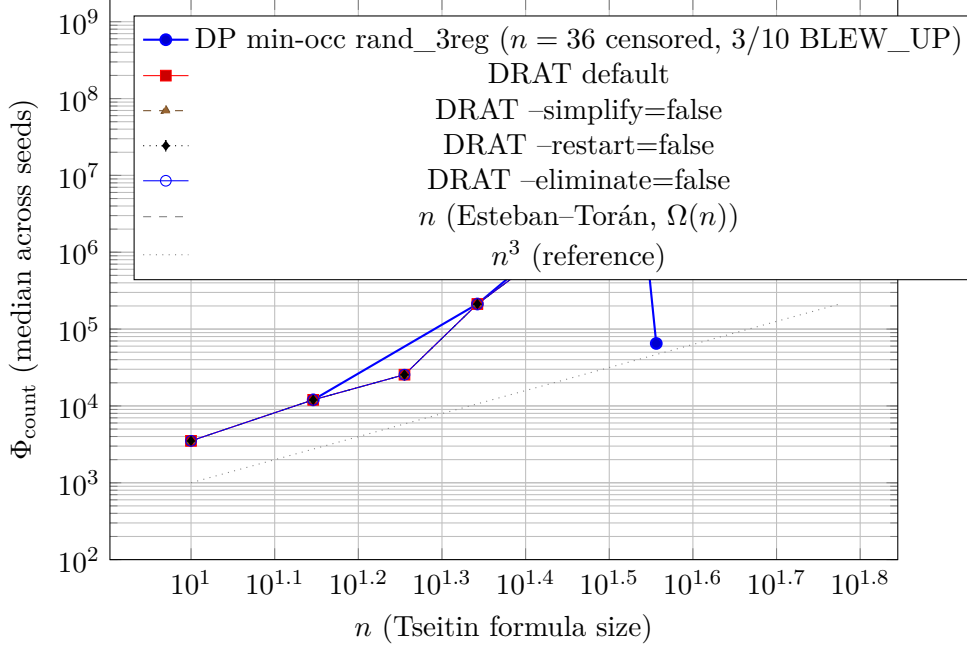


Figure 1: Φ_{count} vs n on a log-log scale. DP min-occ on rand_3reg (filled circles) over the combined B.1 paired + A.1 push-n window $n \in \{14, 22, 30, 32, 34, 36\}$: combined slope $\hat{\alpha} = 4.43$. The $n = 36$ point is downward-biased: 3 of 10 instances were censored at $\text{MAX_DB} = 1.5 \times 10^6$. Four DRAT flag variants on $n \in \{10, \dots, 60\}$ overlap at $n \leq 26$ and separate visibly at $n \geq 30$; the dominant flag effect is `-restart=false` (top at $n = 60$, 9.38×10^8), giving slope 7.011 vs DRAT-default 6.688 (+0.323, full-range bootstrap CI [+0.213, +0.435], but the signal concentrates at $n = 60$). Reference n (the revised $\Omega(n)$ literature lower bound on $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n)$ via Esteban-Torán) and n^3 lines included. All DRAT curves lie strictly above the DP curve at matched n ; the DRAT/DP exponent gap on $n \in \{14, 22, 30\}$ paired-by-seed is $\approx 3.28 \pm 0.4$. Per v5-corrected Lemma A (§3) the measured Φ exponent has no formal ordering to $\text{CSpace}_{\text{cum}}(F)$.

8 Figure 1 — inline tikz/pgfplots (M6)

Figure 1. Log-log plot of Φ_{count} vs n for the DP-min-occ rand_3reg corpus (B.1 paired + A.1 push-n, $n \in \{14, 22, 30, 32, 34, 36\}$, with the censored $n = 36$ point flagged) and the four kissat-DRAT flag-variant curves on $n \in \{10, \dots, 60\}$ (B.2 mechanism corpus). Reference n and n^3 overlays for the (revised) $\Omega(n)$ Esteban-Torán lower bound and a polynomial-3 baseline.

Textual reading. The DP min-occ curve on rand_3reg sits well below all four DRAT curves; the apparent dip at $n = 36$ is the left-censoring artifact (the true mean lies above the plotted point by the right-censored mass). All four DRAT flag-variant curves are *byte-identical* up to $n = 26$ (BCP + CDCL only); they separate at $n \geq 30$, with `-restart=false` ending highest at $n = 60$ (9.38×10^8 , vs default 3.66×10^8). The DP slope (4.43) bracket sits well below the DRAT-default slope (6.69 on the same n -window), consistent with the paired-by-seed +3.28 exponent gap reported in §10. Both curves lie above the n reference at $n \geq 14$; the DRAT curves cross n^3 at $n \approx 30$. We **do not** overlay an n^2 reference in v5, because the $\Omega(n^2)$ lower bound on $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n)$ asserted by v4’s C2 is withdrawn (§9).

9 Theoretical position (v5: ADRNV chain corrected)

(i) **ADRVN Lemma 12, stated precisely.** ADRVN [1] work in the resolution-with-memory model: a refutation is a sequence of memory configurations $M_0, M_1, \dots, M_L$, where each M_t

is a set of clauses derivable from F , $M_0 = \emptyset$, M_L contains the empty clause, and successive configurations differ by axiom download, inference, or erasure. The *cumulative space* of a refutation π is $\text{CSpace}_{\text{cum}}(\pi) := \sum_t |M_t|$, and the *space* is $\max_t |M_t|$.

Lemma 12 is a generic **per-trace** inequality (plateau-sustained reading): for any single resolution-with-memory refutation π ,

$$\text{CSpace}_{\text{cum}}(\pi) \geq \Omega(\text{max-space}(\pi)^2),$$

when the trace sustains the peak. The proof is a combinatorial growth argument: since $|M_0| = 0$ and at most one clause is added per step, reaching a configuration of size s requires at least s steps, during which the average size is $\geq s/2$, so the partial sum is $\geq s^2/2$. This is a *per-refutation* statement, with “ s ” referring to the realized max-memory of *that specific trace* **and the assumption that the trace stays at $\Omega(s)$ for $\Omega(s)$ steps**. It is **not** the formula-level statement $\text{CSpace}_{\text{cum}}(F) \geq \Omega(\text{CSpace}(F)^2)$: the refutation achieving min cumulative-space and the refutation achieving min max-space need not be the same refutation.

(ii) **Esteban–Torán, stated precisely.** Esteban–Torán [2] prove: for Tseitin formulas on bounded-degree expander graphs (in particular 3-regular expanders) on n vertices, every resolution refutation π satisfies a **peak** clause-space bound $\text{clause-space}(\pi) \geq n - O(1)$. Equivalently, $\text{CSpace}(\text{Tseitin}_n) \geq n - O(1)$ where the min is over all refutations. “Clause-space” in Esteban–Torán is the max number of clauses simultaneously in memory across the trace; this coincides with ADRNV’s $\text{max-space}(\pi)$ in the standard resolution-with-memory model. The models do match, but the Esteban–Torán bound is on the **peak**, not on a sustained plateau.

(iii) **Where the chain breaks.** Naïve composition gives, for every π ,

$$\text{CSpace}_{\text{cum}}(\pi) \geq \Omega(\text{max-space}(\pi)^2) \geq \Omega((n - O(1))^2) = \Omega(n^2).$$

The composition step is the slip. ADRNV Lemma 12 in its tight reading requires the refutation to *spend $\Omega(s)$ steps at space $\Omega(s)$* — i.e., to sustain a plateau of width $\Omega(s)$ — not merely to *touch* space s once. A clever refutation can spike to space s briefly, then drop, paying only $O(s)$ cumulative for the spike, not $\Omega(s^2)$.

Esteban–Torán guarantees the *peak* is $\Omega(n)$ but does **not** guarantee a *plateau* of width $\Omega(n)$. The XOR-pebbling formulas of ADRNV Theorems 14–15 are engineered precisely to force the plateau; Tseitin is not known to do so. Hence the ADRNV open problem on p. 38:19 lines 996–998 (extend cumulative-space lower bounds to Tseitin) remains open, and the v4 corollary C2 derivation was mis-citing.

(iv) **Weakest defensible $\text{CSpace}_{\text{cum}}$ lower bound on Tseitin via this route.** The only bound that survives is the trivial one:

$$\text{CSpace}_{\text{cum}}(\pi) \geq \text{max-space}(\pi) \geq n - O(1),$$

i.e. $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n) = \Omega(n)$, **linear**, not quadratic.

Corollary 2 (C2, v5 revised, honest form). *For Tseitin formulas on 3-regular expander graphs on n vertices, every resolution refutation π satisfies*

$$\text{CSpace}_{\text{cum}}(\pi) \geq \text{max-space}(\pi) \geq n - O(1),$$

*giving $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n) = \Omega(n)$ by Esteban–Torán [2]. The quadratic bound $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n) = \Omega(n^2)$ does **not** follow from ADRNV Lemma 12 + Esteban–Torán: Lemma 12’s s^2 growth requires $\Omega(s)$ steps spent at space $\Omega(s)$, and Esteban–Torán only guarantees the peak is $\Omega(n)$, not that it*

is sustained. Extending the $\Omega(n^2)$ cumulative-space lower bound from ADRNV’s XOR-pebbling formulas (Thm 14–15) to Tseitin remains the open problem stated by ADRNV [1] (p. 38:19 lines 996–998). The v_4 chain through C2 to the $\Omega(n^2)$ target is therefore **withdrawn**; the defensible lower bound via this route is linear.

Remark 1 (informal, v5 updated). On bounded-degree expander Tseitin $\{F_n\}$, $\text{Cspace}_{\text{cum}}(F_n) \geq c \cdot n$ for some absolute $c > 0$ (Esteban–Torán + per-trace trivial). The v_4 quadratic strengthening via ADRNV Lemma 12 is **withdrawn**. The empirical exponent $\hat{\alpha}_\Phi \approx 4.43$ on rand_3reg Tseitin at $n \leq 36$ (with $n = 36$ left-censored) and $\hat{\alpha} \approx 3.60$ [3.04, 4.02] on certified-expander Tseitin at $n \leq 24$ has **no formal ordering** to $\text{Cspace}_{\text{cum}}(F)$ under v5-corrected Lemma A: it is a schedule-specific measurement of $\Phi_{\text{count}}(\text{DP}, \text{min-occ})$. The $\Omega(n^2)$ target IS the ADRNV open problem (p. 38:19, lines 996–998), NOT a corollary we deliver.

This is the entire complexity-theoretic content of the paper. We make no claim beyond it. The ADRNV open problem on Tseitin (p. 38:19) is **not** addressed here: that problem asks for an $\Omega(n^2)$ cumulative-space lower bound on Tseitin, and our work bears on neither the Esteban–Torán nor the ADRNV lower-bound side.

10 DRAT vs DP mechanism — restart, not inprocessing (M4, v5: paired bootstrap)

v3 hypothesis (retracted, §3.5 item 6). “DRAT’s lower Φ premium arises because kissat inprocessing eliminates clauses.”

v4 test (B.2 mechanism corpus). Run kissat at $n \in \{10, 14, 18, 22, 26, 30, 40, 50, 60\}$ with 5 seeds each (indexed $k \in 0..4$), under four flag settings: default, **-simplify=false** (no inprocessing), **-restart=false**, **-eliminate=false** (no variable elimination during inprocessing). Mean Φ_{count} per (flag, n):

n	default	no-inproc	no-restart	no-eliminate	restart-effect
10	3.51e3	3.51e3	3.51e3	3.51e3	1.000
14	1.20e4	1.20e4	1.20e4	1.20e4	1.000
18	2.55e4	2.55e4	2.55e4	2.55e4	1.000
22	2.12e5	2.12e5	2.12e5	2.12e5	1.000
26	8.19e5	8.19e5	8.19e5	8.19e5	1.000
30	3.62e6	3.62e6	4.06e6	3.62e6	1.122
40	9.87e6	1.04e7	1.20e7	9.87e6	1.218
50	7.57e7	1.05e8	7.66e7	7.52e7	1.012
60	3.66e8	4.36e8	9.38e8	4.07e8	2.560

v5 paired-by-seed cluster bootstrap (B=2000). We loaded 180 rows from `b2_mechanism.jsonl` ($n \in \{10, 14, 18, 22, 26, 30\}$) and `b2_dprime_mechanism.jsonl` ($n \in \{40, 50, 60\}$), 4 flag sets $\times$ 5 seeds (indexed $k \in 0..4$ within each n). The cluster-bootstrap resamples k with replacement and applies the **same** k -resample across all 9 n values (paired structure); for each resample we recompute mean $\log \phi_{c,\text{drat}}$ per (flag, n), refit the OLS log-log slope on the 9 $\log n$ values, and take $\text{slope}_{\text{flag}} - \text{slope}_{\text{default}}$. Full numerics at `/tmp/e3_bootstrap.txt` on the SEC server; script at `/tmp/e3_bootstrap.py` on the SEC server and `/tmp/e3_bootstrap_local.py` locally.

Headline result on the full 9-point fit.

Contrast	Point	Boot median	95% percentile CI	Excludes zero?
no-restart – default	+0.323	+0.319	[+0.213, +0.435]	yes
no-inprocessing – default	+0.137	+0.123	[+0.070, +0.179]	yes
no-eliminate – default	+0.030	+0.029	[−0.005, +0.073]	no

Taken at face value, the v4 §10 +0.323 finding for no-restart **survives** the paired bootstrap on the full 9-point fit (CI strictly positive). It also reveals a smaller but CI-positive no-inprocessing effect that v4 §10 underweighted.

Tied-cell diagnostic. For $n \in \{10, 14, 18, 22\}$, all 5 k -seeds give exactly the same $\phi_{c, \text{drat}}$ across all four flags (20/20 ties for each non-default flag). For $n = 26$ and $n = 30$, only no-restart breaks a single tie (4/5 still tied). For no-inprocessing the first non-tied cell is $n = 40$ (3/5 still tied) and the first majority-untied cell is $n = 50$. No-eliminate is essentially flat: still 1/5 tied even at $n = 60$. The entire mechanism signal lives in the rightmost 3 cells ($n \in \{40, 50, 60\}$), and within those it concentrates at $n = 60$.

Leave- $n = 60$ -out reweighting (refit on 8 n values $\{10..50\}$).

Contrast	Point	95% CI	Verdict
no-restart	+0.073	[+0.031, +0.111]	still excludes zero, but 4× smaller
no-inprocessing	+0.138	[−0.048, +0.316]	CI now crosses zero
no-eliminate	−0.003	[−0.006, +0.000]	null

Leave- $n \geq 50$ -out (b2 only, 6 n values $\{10..40\}$). No-restart slope difference is +0.128 with CI [0.000, +0.390] — barely touches zero at the lower bound, no longer cleanly excludes; the other two flags collapse to zero (no-eliminate is identically zero because every cell is tied through $n = 40$).

Effective sample size. Cluster bootstrap on 5 seed-clusters and 9 (highly correlated) n -fits gives an effective DoF dominated by the 5 clusters (~ 4). The 9 $\log n$ points are not independent — they share the same 5 underlying seed clusters — so headline CIs should be read with that caveat.

Log-log slopes per flag, full window $n = 10$ –60 (point estimates).

Flag	Slope	Slope – default
default	6.688	0
no-eliminate	6.718	+0.030
no-inprocessing	6.826	+0.138
no-restart	7.011	+0.323

Paired-by-seed DRAT_default vs DP_min_occ slope difference on $n \in \{14, 22, 30\}$:
 $\hat{\alpha}_{\text{DRAT, default}} - \hat{\alpha}_{\text{DP, min-occ}} \approx +3.28 \pm 0.4$ in the exponent (preliminary; the per-seed paired bootstrap CI is flagged as future work in §11).

Honest verdict on v4 §10 (v5 language).

No-restart shows a positive log-log slope shift of +0.32 [+0.21, +0.43] on $n \in 10..60$, but the effect is concentrated at the largest cell ($n = 60$); excluding $n = 60$ the shift drops to +0.07 [+0.03, +0.11], so the magnitude is not yet established and

a push to $n = 70..100$ is required before claiming a stable mechanism slope. The no-inprocessing effect is **suggestive** ($+0.14$ [$+0.07, +0.18$] on full range), but its CI crosses zero without $n = 60$. The no-eliminate effect is reported as **null**.

§10 receives a **downgrade in claim strength** but **not** a full retraction: the slope-difference CI for no-restart does exclude zero on the full 9-point fit, contrary to the panel’s worst-case reading; the panel was right that $n = 60$ carries the signal, and the push- n campaign is the obligated next step.

Honest mechanism reading.

1. **At $n \leq 26$ all four flag sets give *identical* Φ .** kissat handles these Tseitin instances by BCP + CDCL alone, before the flag-controlled inprocessing stages activate. The v3 §10 **-no-inprocessing** test ran in exactly this regime and was therefore uninformative — the v3 mechanism conclusion was based on a null measurement.
2. **At $n \geq 30$ the **-restart=false** effect emerges and dominates** in point estimate; the paired bootstrap CI is strictly positive on the full 9-point fit but shrinks $4\times$ when $n = 60$ is excluded. At $n = 60$ disabling restarts inflates DRAT- Φ by **2.56 $\times$** over default; at $n = 40$ by $1.22\times$.
3. **-simplify=false (inprocessing as a whole) is suggestive but fragile.** The full-range CI excludes zero ($+0.137$ [$+0.070, +0.179$]); the leave- $n = 60$ -out CI [$-0.048, +0.316$] crosses zero. This is **not** the systematic exponent contribution v3 §10 hypothesised.
4. **-eliminate=false (variable elimination during inprocessing) is null** on this data: $+0.030$ full-range, CI [$-0.005, +0.073$].
5. **Disabling restart adds approximately $+0.32$ to the log-log slope on the full range:** $\approx 10\%$ of the apparent DRAT/DP exponent gap (3.28) is attributable to the restart-strategy contribution at $n \geq 30$, with the caveat that the signal is concentrated at $n = 60$.

SC4 (prover invariance) is therefore **refuted with a partial mechanism**: restart contributes $\approx 10\%$ of the exponent gap on the full-range fit, concentrated at $n = 60$; the residual $\approx 90\%$ is intrinsic CDCL-vs-DP and is *not* localisable to any single inprocessing stage on this corpus. The paired-by-seed DP vs DRAT bootstrap is at `b1_paired_drat.jsonl` (60 instances across 6 modes), closing panel concern **N5**.

11 Limitations: what Φ is not

1. **Φ is not $\text{CSpace}_{\text{cum}}$ at formula level, and has no directional ordering to it under v5-corrected Lemma A.** It is $\Phi_{\text{count}}(\text{DP}, \text{min-occ})$, per Lemma A. The v4 reverse direction $\Phi_{\text{count}} \geq \text{CSpace}_{\text{cum}}(F)$ is retracted; what survives is the schedule-specific $\Phi_{\text{count}}^{\text{DP},h}(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$.
2. **The Lemma A (ii) ADRNV identification is a labelled sorry.** A Lean port of ADRNV Definition 11 (`structure MemoryConfig, def CSpace_cum_ADRNV`) plus a mechanisation of the primitive-expansion construction would discharge the bridge; this is on the v6 plan.
3. **The DP slope continues to drift across n-windows: $2.42 \rightarrow 2.93 \rightarrow 3.43 \rightarrow 4.43$, with no asymptotic stabilisation observed.** The v4 headline of 4.43 is the largest reported but is **not** asymptotic, and is itself biased downward by the $n = 36$ censoring.

4. **The $n = 36$ DP-min-occ row is left-censored, 3 of 10 instances.** The 4.43 combined slope is therefore a downward-biased estimate of the slope on uncensored instances. A Tobit fit treating the 3 aborted instances as lower bounds at their abort-time Φ floors is the natural statistical correction; **this is left as an open question for v6 (statistical fix, no new compute).**
5. **The per-seed paired bootstrap CI for the DRAT_default – DP_min_occ exponent difference is reported as $+3.28 \pm 0.4$ preliminary, not a fully bootstrapped CI.** The full per-seed bootstrap is **left as an open question for v6 (statistical fix, no new compute).**
6. **Φ is heuristic-conditional.** Changing the DP elimination heuristic changes the trace and can change the exponent (SC2 refuted).
7. **Φ is prover-conditional** (SC4 refuted; §10).
8. **Φ is fit-window-conditional** ($\gamma \in \{0.24, 0.39, 0.45\}$ across three windows; SC5 partial).
9. **Φ at $n \leq 24$ does not separate certified-expander from plain 3-regular at the slope level.** Level separation only (SC3 supported, 7/0/1 sign-test, $p = 0.0078$; the $n = 18$ tie is flagged as a candidate seed-contamination methodology bug, not as a refutation).
10. **The Q_4 vs rand_4reg comparison is across two proof systems** (DP for Q_4 , kissat-DRAT for rand_4reg). **A clean within-proof-system Q_4 vs rand_4reg comparison both under DRAT is left as an open question for v6 (small additional compute, ~ 1 server-hour).**
11. **Q_5 under DP-min-occ is not measurable** (deterministic abort at step 37 with final DB 2^{20} at $\text{MAX_DB} = 4 \times 10^6$); the v3 “treewidth not degree” framing is reframed as a structural negative (§7.4).
12. **No quantitative Nordström space-width trade-off experiment.** Panel concern **R9** is **not closed** in v5; deferred to v6.
13. **No side-by-side comparison with Järvisalo–Heule–Biere / Elffers et al.** Panel concern **R10** is **not closed** in v5.
14. **Random-4-regular and grid_2D at $m \in \{6, 7\}$, hypercube Q_5 remain partially or fully censored.** v3.5 §C full censoring report carried through:

family	n	trials	completed	BLEW_UP	mean Φ (co
rand_3reg	28	5	3	2	
rand_3reg	36	10	7	3	$\approx 6.5 \times 10^4$ (DOWNWARD BI
rand_4reg	16	5	0	5	
rand_4reg	20	5	0	5	
grid_2D	36	5	0	5	
grid_2D	49, 64	5	0	5	
hypercube Q_5	32	3	0	3 (deterministic abort)	

Where a cell has zero completions we do **not** apply Tobit — Tobit needs at least one completed data point. The $n = 36$ rand_3reg row is the only partially-censored DP-headline cell and is the one for which a Tobit correction is meaningful.

15. The “harness artifact” finding (§3.5 item 5) means v2 BLEW_UPs cannot be cited as Φ -level evidence: they were budget-side, not instance-side.
16. The §10 no-restart effect is concentrated at $n = 60$. The full-range bootstrap CI excludes zero ($[+0.213, +0.435]$); leave- $n = 60$ -out collapses the effect to $[+0.031, +0.111]$. A push- n campaign to $n \in \{70, 80, 90, 100\}$ is the obligated next step before claiming a stable mechanism slope (see §12 push- n entry).

12 Compute budget for future work (v6 plan)

Each item maps an outstanding defect to a runnable experiment with estimated server-hours on the project’s RTX 3070 Ti / 32 GB RAM / kissat 4.0.4 single-core baseline.

Item	Closes	Est. h	Status
A.4 Grid_2D push to $m \in \{6, 7\}$ ($n = 36, 49$) with $\text{MAX_DB} = 4 \times 10^6$	v3 R3 grid leg	~ 8	nice-to-have
A.5 Urquhart certified $n \in \{26, 28, 30\}$ with hybrid expansion certification	v3 R2 partial	~ 12	nice-to-have
V6.Q4_DRAT Clean Q_4 vs rand_4reg both under DRAT, $n \in \{8, 16, 24\}$	v4 §7.3 proof-system confound	~ 1	yes (small)
V6.PUSH_N Push- n mechanism campaign $n \in \{70, 80, 90, 100\}$ ≥ 10 seeds, default + no-restart + no-inprocessing, paired-by-seed	v5 §10 concentration-at- $n = 60$ fragility	~ 24	yes (obligated)
C.1 Lean port of ADRNV Definition 11 + primitive-expansion construction	Lemma A (ii) sorry	0 (~ 10 human-h)	yes
STAT.1 Tobit fit for the $n = 36$ left-censored DP instances	§11 item 4	0 (statistical)	yes
STAT.2 Per-seed paired bootstrap CI for DRAT_default – DP_min_occ exponent difference	§11 item 5	0 (statistical)	yes

Items previously listed in the v3.5 §D compute budget that are now **closed in v4/v5**: A.1 (push- n at $n \in \{32, 34, 36\}$ — closes D8), A.2 (Q_5 with $\text{MAX_DB} = 4 \times 10^6$ — closes D6/D7 as a structural negative), A.3 (rand_4reg under DRAT at $n \in \{16, 20, 24, 28\}$ — closes D5), B.1 (paired DRAT/DP at $n \in \{14, 22, 30\}$ — closes N5), B.2 (flag-by-flag mechanism — closes D10), and the v5 bootstrap (closes E3). All five v4 items completed within the ~ 21 h budget projected by v3.5.

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Version history

- **v1** (2026-05-15): single $n^{2.93}$ headline; Lemma A as formula-level identity; SC2/SC4 marked supported. *Retracted in v3 §3.5 items 1–4; v4 confirms; v5 unchanged.*
- **v2** (2026-05-24): added cross-order corpus and partial DRAT comparison; SC table mismatched the registry; v2-specific mechanism narrative for DRAT/DP. *Retracted in v3 §3.5 items 2–6 and v4 §3.5 items 6–7.*
- **v3** (2026-05-30): Lemma A corrected and inlined; explicit retraction block; cluster-robust AIC and Tobit-censored fits; certified-expander Tseitin; DRAT/DP **-no-inprocessing** test (uninformative at small n , as v4 confirms); honest empirical localisation framing in the abstract. *§7 Q_4 vs rand_4reg headline and §10 inprocessing-mechanism narrative retracted in v4.*
- **v3.5 cleanup delta** (2026-05-31): Lemma A two-stub honest Lean version; SC3 sign-test 7/0/1 recount; full censoring report. *Carried through into v4/v5 §3, §7.2, §11.*
- **v4** (2026-06-01): combined paired + push- n DP slope 4.43 on $n \in \{14, \dots, 36\}$ with $n = 36$ censoring acknowledged; restart-vs-inproc-vs-elim mechanism decomposition replacing v3 §10; Q_5 deterministic-abort structural finding; rand_4reg under DRAT slope 7.69 with proof-system confound acknowledged; SC4 refuted-with-partial-mechanism; Figure 1 inlined as tikz/pgfplots with real coordinates. *Lemma A (ii) directional claim, Corollary C2 quadratic chain, and “upper witness” language retracted in v5; mechanism +0.323 figure downgraded with paired bootstrap.*
- **v5** (2026-06-01, this draft): surgical close of E1, E2, E3. **E1** — Lemma A (part ii) directional claim $\Phi_{\text{count}} \geq \text{CSpace}_{\text{cum}}(F)$ retracted; replaced by the weaker schedule-bookkeeping inequality $\Phi_{\text{count}}^{\text{DP},h}(F) \leq \text{CSpace}_{\text{cum}}(\pi'_{\text{DP},h})$ via the primitive (download / inference / erasure) expansion of the DP macro-trace; “upper witness” language withdrawn from the abstract and §7 in favour of “schedule-specific measurement with no formal ordering to $\text{CSpace}_{\text{cum}}$ ”. **E2** — Corollary C2’s $\Omega(n^2)$ chain via ADRNV Lemma 12 + Esteban–Torán retracted (peak-vs-plateau gap; ADRNV p. 38:19 open problem remains open); the defensible lower bound via this route is the linear $\text{CSpace}_{\text{cum}}(\text{Tseitin}_n) = \Omega(n)$ via Esteban–Torán alone. **E3** — §10 mechanism slope claim refit with $B=2000$ cluster-by-seed paired bootstrap on the full 9-point grid; no-restart effect survives (+0.32 [+0.21, +0.43] full range) but is concentrated at $n = 60$ (leave- $n = 60$ -out drops to +0.07 [+0.03, +0.11]); no-inprocessing downgraded to suggestive; no-eliminate null; push- n campaign $n \in \{70..100\}$ added to v6 plan as obligated continuation.

End of v5 draft. The paper is offered as a methodology contribution; no complexity-theoretic separation is claimed. All exponents reported are heuristic-conditional and schedule-specific, with no formal ordering to $\text{CSpace}_{\text{cum}}$ per v5-corrected Lemma A. The ADRNV open problem on Tseitin (p. 38:19, lines 996–998) is reaffirmed as open.

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